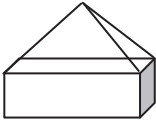


Ref	Content descriptor	Concepts and skills
GM z	Find circumferences and areas of circles	- Find circumferences of circles and areas enclosed by circles - Recall and use the formulae for the circumference of a circle and the area enclosed by a circle - Use $\pi \approx 3.142$ or use the π button on a calculator - Find the perimeters and areas of semicircles and quarter circles Calculate the lengths of arcs and the areas of sectors of circles Answers in terms of π may be required - Find the surface area of a cylinder
GM aa	Calculate volumes of right prisms and shapes made from cubes and cuboids	- Calculate volumes of right prisms, including the triangular prism, and shapes made from cubes and cuboids - Recall and use the formula for the volume of a cuboid - Find the volume of a cylinder - Use volume to solve problems
GM bb	Solve mensuration problems involving more complex shapes and solids	Solve problems involving more complex shapes and solids, including segments of circles and frustums of cones Find the surface area and volumes of compound solids constructed from cubes, cuboids, cones, pyramids, spheres, hemispheres, cylinders Examples:    Solve problems including examples of solids in everyday use Find the area of a segment of a circle given the radius and length of the chord

Ref	Content descriptor	Concepts and skills
GM cc	Use vectors to solve problems	• Understand and use vector notation • Calculate, and represent graphically, the sum of two vectors, the difference of two vectors and a scalar multiple of a vector • Calculate the resultant of two vectors • Solve geometrical problems in 2-D using vector methods • Apply vector methods for simple geometrical proofs

4 Measures

What students need to learn:

Ref	Content descriptor	Concepts and skills
GM m	Use and interpret maps and scale drawings	• Use and interpret maps and scale drawings • Read and construct scale drawings • Draw lines and shapes to scale • Estimate lengths using a scale diagram
GM n	Understand and use the effect of enlargement for perimeter, area and volume of shapes and solids	• Understand the effect of enlargement for perimeter, area and volume of shapes and solids • Understand that enlargement does not have the same effect on area and volume • Use simple examples of the relationship between enlargement and areas and volumes of simple shapes and solids • Use the effect of enlargement on areas and volumes of shapes and solids • Know the relationships between linear, area and volume scale factors of mathematically similar shapes and solids
GM o	Interpret scales on a range of measuring instruments and recognise the inaccuracy of measurements	• Know that measurements using real numbers depend upon the choice of unit • Recognise that measurements given to the nearest whole unit may be inaccurate by up to one half in either direction

Ref	Content descriptor	Concepts and skills												
GM p	Convert measurements from one unit to another	- Convert between units of measure in the same system (NB: Conversion between imperial units will be given. Metric equivalents should be known) - Know rough metric equivalents of pounds, feet, miles, pints and gallons: <table><thead><tr><th>Metric</th><th>Imperial</th></tr></thead><tbody><tr><td>1 kg</td><td>2.2 pounds</td></tr><tr><td>1 l</td><td>1 $\frac{3}{4}$ pints</td></tr><tr><td>4.5 l</td><td>1 gallon</td></tr><tr><td>8 km</td><td>5 miles</td></tr><tr><td>30 cm</td><td>1 foot</td></tr></tbody></table> - Convert between imperial and metric measures - Convert between metric area measures - Convert between metric volume measures - Convert between metric speed measures - Convert between metric units of volume and units of capacity measures, eg 1 m³ = 1000 cm³	Metric	Imperial	1 kg	2.2 pounds	1 l	1 $\frac{3}{4}$ pints	4.5 l	1 gallon	8 km	5 miles	30 cm	1 foot
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4.5 l	1 gallon													
8 km	5 miles													
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GM q	Make sensible estimates of a range of measures	- Make sensible estimates of a range of measures in everyday settings - Choose appropriate units for estimating or carrying out measurements												
GM r	Understand and use bearings	- Use three-figure bearings to specify direction - Mark on a diagram the position of the point <i>B</i> given its bearing from point <i>A</i> - Measure or draw a bearing between the points on a map or scaled plan - Given the bearing of a point <i>A</i> from point <i>B</i>, work out the bearing of <i>B</i> from <i>A</i>												
GM s	Understand and use compound measures	Understand and use compound measures, including speed and density												
GM t	Measure and draw lines and angles	- Measure and draw lines, to the nearest mm - Measure and draw angles, to the nearest degree												

Ref	Content descriptor	Concepts and skills
GM u	Draw triangles and other 2-D shapes using ruler and protractor	• Make accurate drawing of triangles and other 2-D shapes using a ruler and a protractor • Make an accurate scale drawing from a diagram • Use accurate drawing to solve a bearings problem